

Volatility

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}$$

Parameter set

$$\boldsymbol{\theta} = \{\beta_0, \beta_1, \beta_2, \theta_1, \theta_2, \lambda_{1,0}, \lambda_{1,1}, \lambda_{2,0}, \lambda_{2,1}\}$$

SDE form

$$d\mathbf{X}_t = \mathbf{F}(\mathbf{X}_t; \boldsymbol{\theta}) dt + \boldsymbol{\Sigma}(\mathbf{X}_t; \boldsymbol{\theta}) dW_t$$

$$d \begin{pmatrix} S_t \\ R_{1,t} \\ R_{1,0,t} \\ R_{2,t} \\ R_{2,0,t} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ (1-\theta_2)\lambda_{2,0}(\sigma_t^2 - R_{2,0,t}) + \theta_2\lambda_{2,1}(\sigma_t^2 - R_{2,1,t}) \\ \lambda_{2,0}\sigma_t^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix}}_{\mathbf{F}} dt + \underbrace{\begin{pmatrix} \sigma_t S_t \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})\sigma_t \\ \lambda_{1,0}\sigma_t \\ 0 \\ 0 \end{pmatrix}}_{\boldsymbol{\Sigma}} dW_t$$

 $\mathbf{F}$ function:

$$\mathbf{F} = \begin{pmatrix} 0 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ (1-\theta_2)\lambda_{2,0}(\sigma_t^2 - R_{2,0,t}) + \theta_2\lambda_{2,1}(\sigma_t^2 - R_{2,1,t}) \\ \lambda_{2,0}\sigma_t^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ (1-\theta_2)\lambda_{2,0}((\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})^2 - R_{2,0,t}) + \theta_2\lambda_{2,1}((\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})^2 - R_{2,1,t}) \\ \lambda_{2,0}\sigma_t^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix}$$

 $\boldsymbol{\Sigma}$ function:

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_t S_t \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})\sigma_t \\ \lambda_{1,0}\sigma_t \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} (\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})S_t \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}) \\ \lambda_{1,0}(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}) \\ 0 \\ 0 \end{pmatrix}$$

Another F function:

$$d \begin{pmatrix} X_t^{(1)} \\ X_t^{(2)} \\ X_t^{(3)} \\ X_t^{(4)} \\ X_t^{(5)} \end{pmatrix} = \underbrace{\begin{pmatrix} \dots \end{pmatrix}}_F dt + \underbrace{\begin{pmatrix} \dots \end{pmatrix}}_\Sigma dW_t$$

where $X_t^{(1)} = \log S_t$, $X_t^{(2)} = R_{1,t}$, $X_t^{(3)} = R_{1,0,t}$, $X_t^{(4)} = \sqrt{R_{2,t}}$, $X_t^{(5)} = R_{2,0,t}$

First entry

Itô:

$$\begin{aligned} dX_t^{(1)} &= d \log S_t, \quad dS_t = S_t \sigma_t dW_t, \quad \frac{d}{dS_t} \log(S_t) = \frac{1}{S_t}, \quad \frac{d^2}{dS_t^2} \log(S_t) = -\frac{1}{S_t^2} \\ dX_t^{(1)} &= -\frac{1}{2} \sigma_t^2 \frac{1}{S_t^2} S_t^2 dt + \sigma_t \frac{1}{S_t} S_t dW_t = -\frac{1}{2} \sigma_t^2 dt + \sigma_t dW_t \end{aligned}$$

Second entry

$$\begin{aligned} dX_t^{(2)} &= dR_{1,t} \\ &= (1 - \theta_1) dR_{1,0,t} + \theta_1 dR_{1,1,t} \\ &= (1 - \theta_1) \lambda_{1,0} (\sigma_t dW_t - R_{1,0,t} dt) + \theta_1 \lambda_{1,1} (\sigma_t dW_t - R_{1,1,t} dt) \\ &= ((1 - \theta_1) \lambda_{1,0} + \theta_1 \lambda_{1,1}) \sigma_t dW_t - ((1 - \theta_1) \lambda_{1,0} R_{1,0,t} + \theta_1 \lambda_{1,1} R_{1,1,t}) dt \end{aligned}$$

Third entry

$$dX_t^{(3)} = dR_{1,0,t} = \lambda_{1,0} (\sigma_t dW_t - R_{1,0,t} dt)$$

Fourth entry

$$\begin{aligned} dX_t^{(4)} &= d\sqrt{R_{2,t}} \\ &= \frac{1}{2\sqrt{R_{2,t}}} dR_{2,t} - \frac{1}{8R_{2,t}^{3/2}} (dR_{2,t})^2 \\ &= \frac{1}{2\sqrt{R_{2,t}}} (((1 - \theta_2) \lambda_{2,0} + \theta_2 \lambda_{2,1}) \sigma_t dW_t - ((1 - \theta_2) \lambda_{2,0} R_{2,0,t} + \theta_2 \lambda_{2,1} R_{2,1,t}) dt) \\ &\quad - \frac{1}{8R_{2,t}^{3/2}} ((1 - \theta_2) \lambda_{2,0} + \theta_2 \lambda_{2,1})^2 \sigma_t^2 dt \\ &= \left(-\frac{1}{2\sqrt{R_{2,t}}} ((1 - \theta_2) \lambda_{2,0} R_{2,0,t} + \theta_2 \lambda_{2,1} R_{2,1,t}) - \frac{1}{8R_{2,t}^{3/2}} ((1 - \theta_2) \lambda_{2,0} + \theta_2 \lambda_{2,1})^2 \sigma_t^2 \right) dt \\ &\quad + \frac{1}{2\sqrt{R_{2,t}}} ((1 - \theta_2) \lambda_{2,0} + \theta_2 \lambda_{2,1}) \sigma_t dW_t \end{aligned}$$

Where we have used that $dR_{2,t} = ((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})\sigma_t dW_t - ((1-\theta_2)\lambda_{2,0}R_{2,0,t} + \theta_2\lambda_{2,1}R_{2,1,t})dt$ and $(dR_{2,t})^2 = ((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})^2\sigma_t^2 dt$

Fifth and last entry

$$dX_t^{(5)} = dR_{2,0,t} = \lambda_{2,0}(\sigma_t^2 - R_{2,0,t})dt$$

Matrix form

$$d \begin{pmatrix} \log(S_t) \\ R_{1,t} \\ R_{1,0,t} \\ \sqrt{R_{2,t}} \\ R_{2,0,t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}\sigma_t^2 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ -\frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0}R_{2,0,t} + \theta_2\lambda_{2,1}R_{2,1,t}) - \frac{1}{8R_{2,t}^{3/2}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})^2\sigma_t^2 \\ \lambda_{2,0}\sigma_t^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix} dt + \begin{pmatrix} \sigma_t \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})\sigma_t \\ \lambda_{1,0}\sigma_t \\ \frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})\sigma_t \\ 0 \end{pmatrix} dW_t$$

$\mathbf{F}$ function:

$$\mathbf{F} = \begin{pmatrix} -\frac{1}{2}\sigma_t^2 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ -\frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0}R_{2,0,t} + \theta_2\lambda_{2,1}R_{2,1,t}) - \frac{1}{8R_{2,t}^{3/2}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})^2\sigma_t^2 \\ \lambda_{2,0}\sigma_t^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})^2 \\ -(1-\theta_1)\lambda_{1,0}R_{1,0,t} - \theta_1\lambda_{1,1}R_{1,1,t} \\ -\lambda_{1,0}R_{1,0,t} \\ -\frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0}R_{2,0,t} + \theta_2\lambda_{2,1}R_{2,1,t}) - \frac{1}{8R_{2,t}^{3/2}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})^2(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})^2 \\ \lambda_{2,0}(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}})^2 - \lambda_{2,0}R_{2,0,t} \end{pmatrix}$$

$\mathbf{\Sigma}$ function:

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_t \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})\sigma_t \\ \lambda_{1,0}\sigma_t \\ \frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})\sigma_t \\ 0 \end{pmatrix} = \begin{pmatrix} \beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}} \\ ((1-\theta_1)\lambda_{1,0} + \theta_1\lambda_{1,1})(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}) \\ \lambda_{1,0}(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}) \\ \frac{1}{2\sqrt{R_{2,t}}}((1-\theta_2)\lambda_{2,0} + \theta_2\lambda_{2,1})(\beta_0 + \beta_1 R_{1,t} + \beta_2 \sqrt{R_{2,t}}) \\ 0 \end{pmatrix}$$